

# Tute (Week 5)

## Entailment

### Question 1

Is the following claim

$$\text{If } \phi \not\models \psi \text{ then } \phi \models \neg\psi$$

1. a true statement about propositional formulas?
2. a true statement about predicate formulas?

1. No it is not true, consider for example  $\phi = p$  and  $\psi = q$  (where  $p$  and  $q$  are propositional variables). Then neither  $P \models Q$  nor  $P \models \neg Q$ .
2. Not true either. For example, let  $\phi = (x = y)$  and  $\psi = (z = w)$ .

### Question 2

Show that the following is *not* valid:

$$\forall x(P(x) \rightarrow Q(x)) \models \exists xQ(x)$$

We need to find a model where the first formula holds but the second does not. Consider a model  $\Box$  where the domain has one element, where  $P^\Box = \emptyset$  and  $Q^\Box = \emptyset$ . Then  $\llbracket \forall x(P(x) \rightarrow Q(x)) \rrbracket_\Box$  holds (the implication becomes vacuously true). But  $\llbracket \exists xQ(x) \rrbracket_\Box$  does not.

### Question 3

Show that the following is valid:

$$\forall xP(x) \models \exists xP(x)$$

We can use natural deduction:

$$\frac{\frac{\forall x P(x)}{P(c)} \forall\text{-E}}{\exists x P(x)} \exists\text{-I}$$

We could also try a direct semantic argument, but this is considerably more involved.

## Limitations of first-order logic

### Question 1

We can use predicate logic to talk about almost anything, by choosing an appropriate vocabulary. But we can't use it to talk about nothing, because we require a non-empty domain of discourse.

Why do you think this restriction exists?

Natural deduction becomes unsound for an empty domain of discourse. For example, consider  $\forall\text{-E}$ :

$$\frac{\forall x P(x)}{P(c)}$$

If the domain of discourse is empty, the premise of this rule is vacuously true: everything in the domain of discourse does indeed satisfy  $P$ . But the conclusion is false (or ill-formed, if you prefer):  $P(c)$  cannot hold because there is no  $c$ .

You can use similar arguments to show breakdowns of useful logical laws. For example, the law

$$\exists x. \top \equiv \top$$

would fail in an empty domain of discourse.

### Question 2

Recall the Peano axioms from the W3 tute. Can we do Peano arithmetic in first-order predicate logic? First, let our vocabulary include the constant symbol  $Z$  and the unary function symbol  $S$ .

Let  $\phi$  be a first-order predicate formula with this vocabulary.

1. Suppose it is the case that

$$\models \phi$$

Does it follow that  $\phi$  is true in Peano arithmetic?

2. Suppose it is the case that

$$\not\models \phi$$

Does it follow that  $\phi$  is false in Peano arithmetic?

3. Suppose I want to use FOL + the Peano axioms to prove  $\phi$ . Let's try to make a theory  $\square$  that includes the Peano axioms, so that we can prove

$$\square \models \phi$$

Write down a set of formulas that capture all but the last Peano axiom.

4. For the last axiom, it unfortunately turns out that we can't write down a FOL formula that captures induction on arbitrary propositions such as  $\phi$ . Why not? Are there any workarounds?

1. Yes. Peano arithmetic is one possible model of our vocabulary, and  $\models \phi$  means that  $\phi$  is true in every possible model (and environment).
2. No.  $\not\models \phi$  means that  $\phi$  is false in *at least one* model, but it may still be true in Peano arithmetic. For example, the formula  $\neg(S(Z) = Z)$  is false in Peano arithmetic, but true in any model with a one-element domain of discourse.
3. The first two Peano axioms are more syntax than semantics, and hence do not require any FOL axioms. For example, "there is a number  $Z$ " follows immediately from the fact that  $Z$  is in the vocabulary. For axioms 3 and 4 we can do the following:

$$\{\forall n(\neg(S(n) = Z)), \\ \forall n\forall m(S(n) = S(m) \rightarrow n = m)\}$$

4. Using logical notation instead of set notation, the induction axiom can be stated as:

$$\forall P((P(Z) \wedge \forall n(P(n) \rightarrow P(S(n)))) \rightarrow \forall n(P(n)))$$

Unfortunately, the above is not a wff in first-order predicate logic! That's because first-order logic only allows quantification over (term) variables.  $P$ , however, is not a term variable but a predicate symbol.

What we can do is add an infinite family of *instances* of the induction axiom to our theory. The idea is that for every wff  $\phi(n)$ , we add an axiom

$$(\phi(Z) \wedge \forall n(\phi(n) \rightarrow \phi(S(n)))) \rightarrow \forall n(\phi(n))$$

This approach allows us to do Peano arithmetic for most practical purposes, but will still not uniquely characterise Peano arithmetic: there will be other models that also satisfy these axioms. This is a bit beyond the scope of the course, but if you're interested in why, look up the compactness theorem and non-standard models of arithmetic. One place to start is [here](#).

## Predicate Natural Deduction

Give Natural Deduction proofs of the following:

1.  $\vdash \forall x F(x) \vee \neg \forall x F(x)$
2.  $\vdash \forall x \exists y (Q(y) \rightarrow Q(x))$
3.  $(m = n) \vee (n = o), A(n) \vdash A(m) \vee A(o)$
4.  $\exists x J(x), \exists x \neg J(x) \vdash \exists x \exists y \neg (x = y)$

1. This proof uses only the rules from the propositional fragment of ND (along with the derived rule LEM, which is not needed but simplifies matters)

$$\frac{\frac{[\forall x F(x)]}{\forall x F(x) \vee \neg \forall x F(x)} \vee\text{-I} \quad \frac{[\neg \forall x F(x)]}{\forall x F(x) \vee \neg \forall x F(x)} \vee\text{-I}}{\forall x F(x) \vee \neg \forall x F(x)} \text{LEM}$$

$$\begin{array}{c} 2. \\ \frac{[Q(c)]}{Q(c) \rightarrow Q(c)} \rightarrow\text{-I} \\ \frac{Q(c) \rightarrow Q(c)}{\exists y (Q(y) \rightarrow Q(c))} \exists\text{-I} \\ \frac{\exists y (Q(y) \rightarrow Q(c))}{\forall x \exists y (Q(y) \rightarrow Q(x))} \forall\text{-I} \end{array}$$

$$\begin{array}{c} 3. \\ \frac{(m = n) \vee (n = o) \quad \frac{\frac{[m = n] \quad A(n)}{A(m)} =\text{-E} \quad \frac{[n = o] \quad A(n)}{A(o)} =\text{-E}}{A(m) \vee A(o)} \vee\text{-I} \quad \frac{A(m) \vee A(o)}{A(m) \vee A(o)} \vee\text{-I}}{A(m) \vee A(o)} \vee\text{-E} \end{array}$$

$$\begin{array}{c} 4. \\ \frac{\exists x J(x) \quad \frac{\frac{[\neg J(d)] \quad \frac{[c = d] \quad [J(c)]}{J(d)} =\text{-E}}{\neg J(d)} \neg\text{-E}}{\perp} \neg\text{-I} \quad \frac{\perp}{\neg(c = d)} \neg\text{-I} \quad \frac{\neg(c = d)}{\exists y \neg(c = y)} \exists\text{-I} \quad \frac{\exists y \neg(c = y)}{\exists x \exists y \neg(x = y)} \exists\text{-I} \quad \frac{\exists x \neg J(x) \quad \exists x \exists y \neg(x = y)}{\exists x \exists y \neg(x = y)} \exists\text{-E} \quad \frac{\exists x J(x) \quad \exists x \exists y \neg(x = y)}{\exists x \exists y \neg(x = y)} \exists\text{-E} \end{array}$$

